Purple Lab detection and response validation

Controlled tests in Microsoft Sentinel

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Results overview

Process and security evidence, Sentinel incidents and automation execution were confirmed for the controlled tests. This update covers the QuickBuck test, local triage and final protection checks. Sysmon ingestion into Sentinel and detection of bulk encryption remain unconfirmed.

Section	Confirmed result	Incident
Test 01 Discovery	Process telemetry and automation	#94 Medium
Test 02 Defense Evasion	Archive masquerading and five correlated indicators	#98 Medium #105 High
Test 03 Cobalt LAB	Three indicators and 15 related events	#117 and #119 High
Test 04 WMI	Local registration, control test and event 4688 in Sentinel	Detection pending validation
Test 05 QuickBuck	AV blocking, completed test and local Sysmon evidence	#413 and #420 High AV

Environment and method

The target was WIN10-NIKOLA.b33keep3r.lab, IPv4 10.0.2.7. Kali Linux at 10.0.2.8 served as the HTTP receiver for Cobalt LAB on port 8028. Security events are collected through Azure Monitor Agent and DCR configuration in the purplelab-sentinel workspace.

We checked local evidence, telemetry ingestion, KQL detections, incidents and playbook comments. Defender remained enabled; the completed QuickBuck test used a temporary exclusion for one EXE. Execution conditions are stated alongside the results.

Interpreting the findings

The tests ran in a lab environment using test artifacts. The findings distinguish execution, blocked attempts and unconfirmed outcomes. Multiple incidents may result from reprocessing the same events.

Times in the text are UTC. The portal and consoles in the screenshots may show Europe/Belgrade local time, which was UTC+1 during this session. Incident counts do not represent independent test executions.

Test 01 Discovery

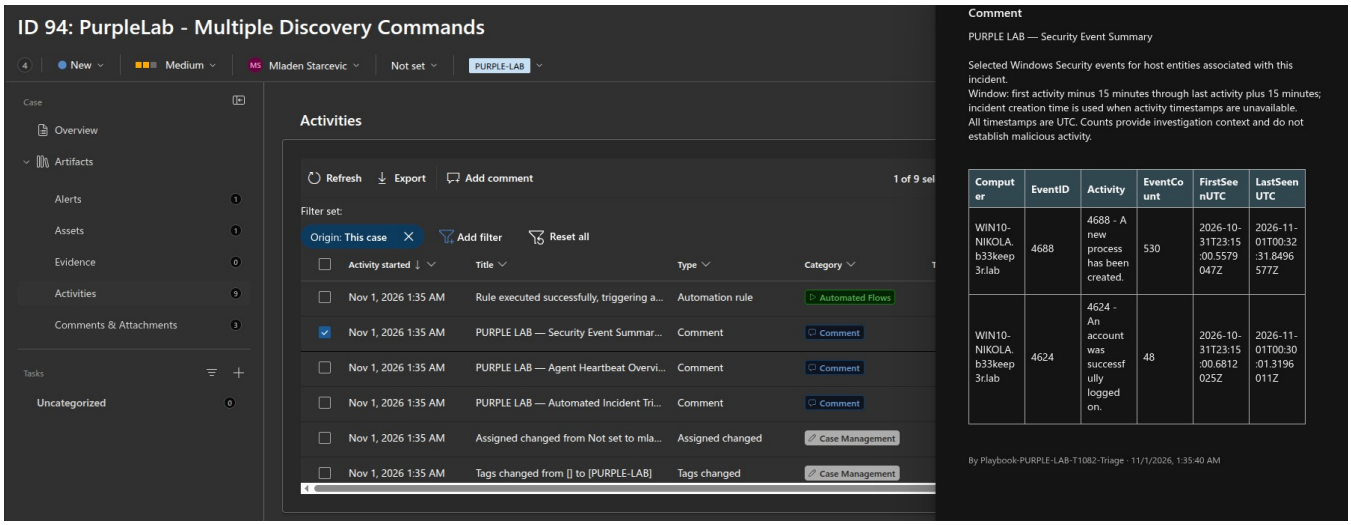
MITRE ATT&CK T1082 T1033 T1069.001

The aim was to check visibility of common host discovery activity. The test ran nbtscan, whoami, systeminfo, net localgroup administrators and WMIC queries for patches and shares.

Results | Chart | Add bookmark

TimeGenerated [UTC]	Computer	NewProcessName	CommandLine	ParentProcessName	IngestedUTC [UTC]
10/31/2026, 11:03:10.662 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\cmd.exe	[redacted]	C:\Windows\System32\WindowsPowerShell\v1.0\powershell...	10/31/2026, 11:30:08.2
10/31/2026, 11:03:22.540 PM	WIN10-NIKOLA.b33keep3r.lab	C:\TMP\nbtscan.exe	"C:\TMP\nbtscan.exe" 192.168.1.0/24	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:09.5
10/31/2026, 11:03:30.561 PM	WIN10-NIKOLA.b33keep3r.lab	C:\TMP\nbtscan.exe	"C:\TMP\nbtscan.exe" 10.10.0.0/24	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:09.5
10/31/2026, 11:03:40.540 PM	WIN10-NIKOLA.b33keep3r.lab	C:\TMP\nbtscan.exe	"C:\TMP\nbtscan.exe" 172.28.29.0/24	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:18.8
10/31/2026, 11:03:52.844 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\whoami.exe	whoami	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:18.8
10/31/2026, 11:03:52.895 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\systeminfo.exe	systeminfo	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:18.8
10/31/2026, 11:03:54.906 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\net.exe	net localgroup administrators	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:18.8
10/31/2026, 11:03:54.920 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\net1.exe	C:\Windows\system32\net1 localgroup administrators	C:\Windows\System32\net.exe	10/31/2026, 11:30:18.8
10/31/2026, 11:03:54.955 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\wbem\WMIC.e...	wmic qfe list full	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:18.8
10/31/2026, 11:03:55.234 PM	WIN10-NIKOLA.b33keep3r.lab	C:\Windows\System32\wbem\WMIC.e...	wmic share get	C:\Windows\System32\cmd.exe	10/31/2026, 11:30:18.8

E01 Sentinel event 4688 process evidence for the Discovery test on WIN10-NIKOLA. The test tool name is omitted in the marked fields.
Source: image(20261031-233251).png | cropped from original | redacted



E02 Incident 94 and Activities show a successful automation rule, owner assignment, a tag and three playbook comments.
Source: image(20261101-003734).png

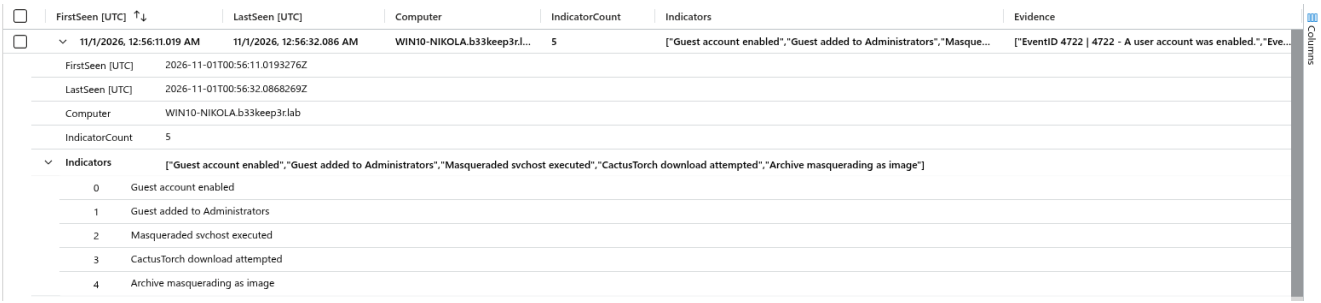
Result: execution telemetry was confirmed, and the PurpleLab - Multiple Discovery Commands rule generated a Medium incident. Incident 94 includes Security Event Summary, Agent Heartbeat Overview and Automated Incident Triage.

Limitation: nbtscan used the default ranges 192.168.1.0/24, 10.10.0.0/24 and 172.28.29.0/24. Timeouts and inserted sample output do not establish discovery of lab devices. The batch also reported the error "+1 was unexpected at this time".

Test 02 Defense Evasion

MITRE ATT&CK T1036.005 T1036.008 T1098.007

The tests covered system file masquerading, an archive with a JPG extension and changes to the Guest account. The multistage correlation also included an attempted CactusTorch dropper download. This section groups related tests; some actions belong to other ATT&CK tactics.

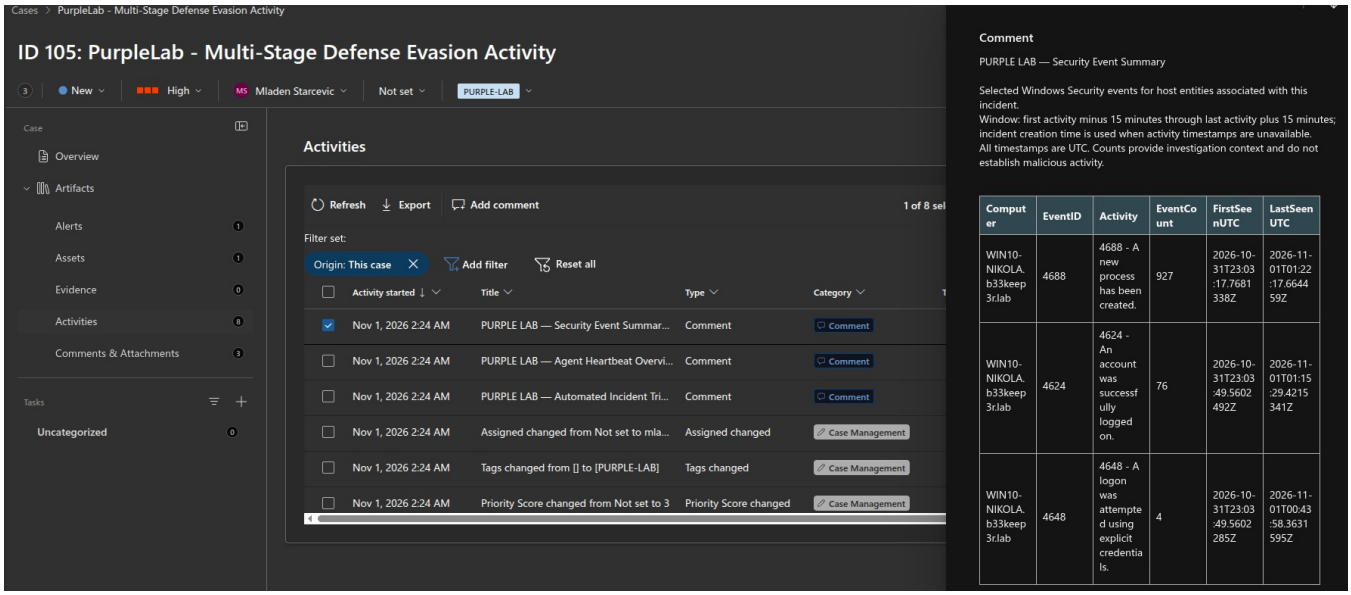


FirstSeen [UTC]	LastSeen [UTC]	Computer	IndicatorCount	Indicators	Evidence
11/1/2026, 12:56:11.019 AM	11/1/2026, 12:56:32.086 AM	WIN10-NIKOLA.b33keep3r.l...	5	["Guest account enabled","Guest added to Administrators","Masque...	["EventID 4722 | 4722 - A user account was enabled.","Eve...

FirstSeen [UTC]	2026-11-01T00:56:11.0193276Z
LastSeen [UTC]	2026-11-01T00:56:32.0868269Z
Computer	WIN10-NIKOLA.b33keep3r.lab
IndicatorCount	5
Indicators	["Guest account enabled","Guest added to Administrators","Masqueraded svchost executed","CactusTorch download attempted","Archive masquerading as image"]

0	Guest account enabled
1	Guest added to Administrators
2	Masqueraded svchost executed
3	CactusTorch download attempted
4	Archive masquerading as image

E03 Five indicators between 00:56:11 and 00:56:32 UTC, including Guest changes, a fake svchost, an attempted download and a disguised archive.
Source: image(20261101-011830).png | cropped from original



Cases > PurpleLab - Multi-Stage Defense Evasion Activity

ID 105: PurpleLab - Multi-Stage Defense Evasion Activity

New | High | MS Mladen Starcevic | Not set | PURPLE-LAB

Case
Overview
Artifacts
Alerts
Assets
Evidence
Activities
Comments & Attachments
Tasks
Uncategorized

Activities

Refresh | Export | Add comment | 1 of 8 sel

Filter set:
Origin: This case | Add filter | Reset all

Activity started	Title	Type	Category
Nov 1, 2026 2:24 AM	PURPLE LAB — Security Event Summar...	Comment	Comment
Nov 1, 2026 2:24 AM	PURPLE LAB — Agent Heartbeat Overvi...	Comment	Comment
Nov 1, 2026 2:24 AM	PURPLE LAB — Automated Incident Tri...	Comment	Comment
Nov 1, 2026 2:24 AM	Assigned changed from Not set to mla...	Assigned changed	Case Management
Nov 1, 2026 2:24 AM	Tags changed from [] to [PURPLE-LAB]	Tags changed	Case Management
Nov 1, 2026 2:24 AM	Priority Score changed from Not set to 3	Priority Score changed	Case Management

Comment

PURPLE LAB — Security Event Summary

Selected Windows Security events for host entities associated with this incident.
Window: first activity minus 15 minutes through last activity plus 15 minutes; incident creation time is used when activity timestamps are unavailable.
All timestamps are UTC. Counts provide investigation context and do not establish malicious activity.

Computer	EventID	Activity	EventCount	FirstSeenUTC	LastSeenUTC
WIN10-NIKOLA.b33keep3r.lab	4688	4688 - A new process has been created.	927	2026-10-31T23:03:17.7681338Z	2026-11-01T01:22:17.664459Z
WIN10-NIKOLA.b33keep3r.lab	4624	4624 - An account was successfully logged on.	76	2026-10-31T23:03:49.5602492Z	2026-11-01T01:15:29.4215341Z
WIN10-NIKOLA.b33keep3r.lab	4648	4648 - A logon was attempted using explicit credentials.	4	2026-10-31T23:03:49.5602285Z	2026-11-01T00:43:58.3631595Z

E04 High incident 105 with an owner, the PURPLE-LAB tag and three playbook comments, including Security Event Summary.
Source: image(20261101-012555).png

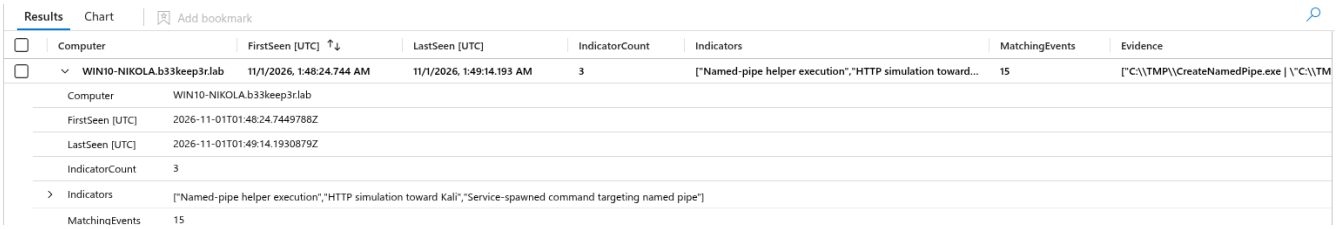
Result: Guest enablement (4722), addition to Administrators (4732) and execution of C:\Users\Public\svchost.exe were confirmed. An earlier standalone test of the s01.jpg archive generated Medium incident 98. Events 4733 and 4725 support Guest cleanup.

Limitation: the CactusTorch download was an attempt; a successful shell or injection was not confirmed. Hosts file changes and reg import produced errors. Rules in the historical screenshots use broader mappings; here, T1036.005 and T1036.008 refer to masquerading, and T1098.007 to adding an existing account to a group.

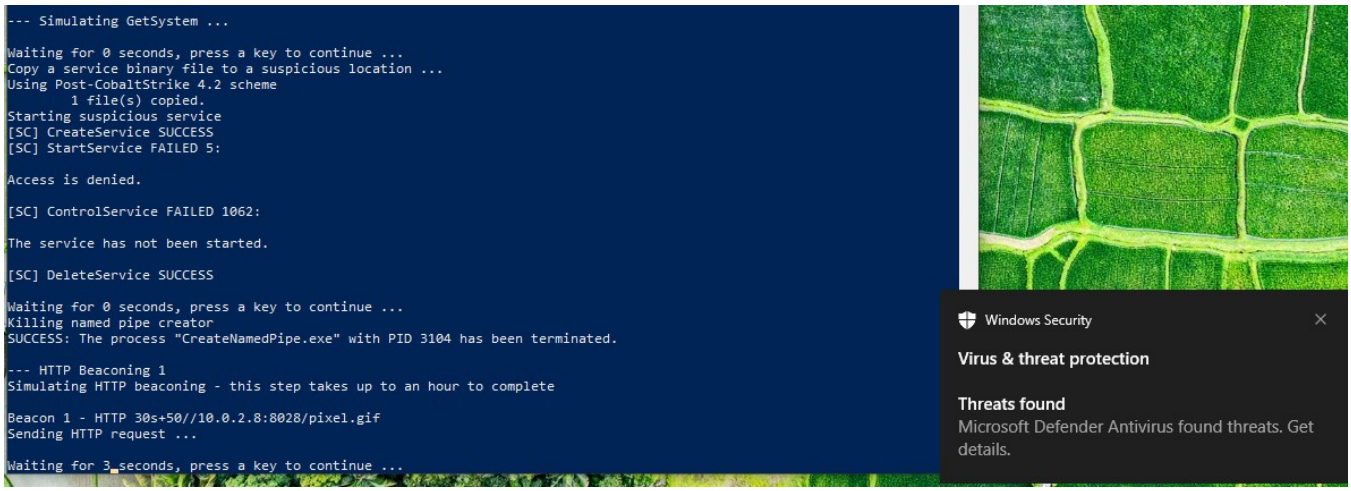
Test 03 Cobalt LAB behavior

MITRE ATT&CK T1569.002 T1059.003 T1071.001

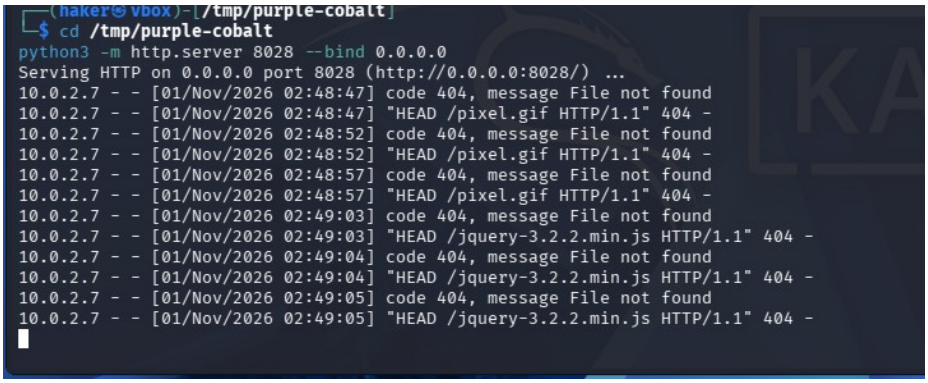
A modified LAB copy of the module simulated named-pipe patterns, a temporary service and HTTP beaconing to 10.0.2.8:8028. Execution ran from 01:48:21 to 01:49:19 UTC in a controlled test environment, without an actual Beacon or Team Server system.



E05 Correlation confirms three indicators and 15 events between 01:48:24 and 01:49:14 UTC.
Source: image(20261101-020520).png | cropped from original



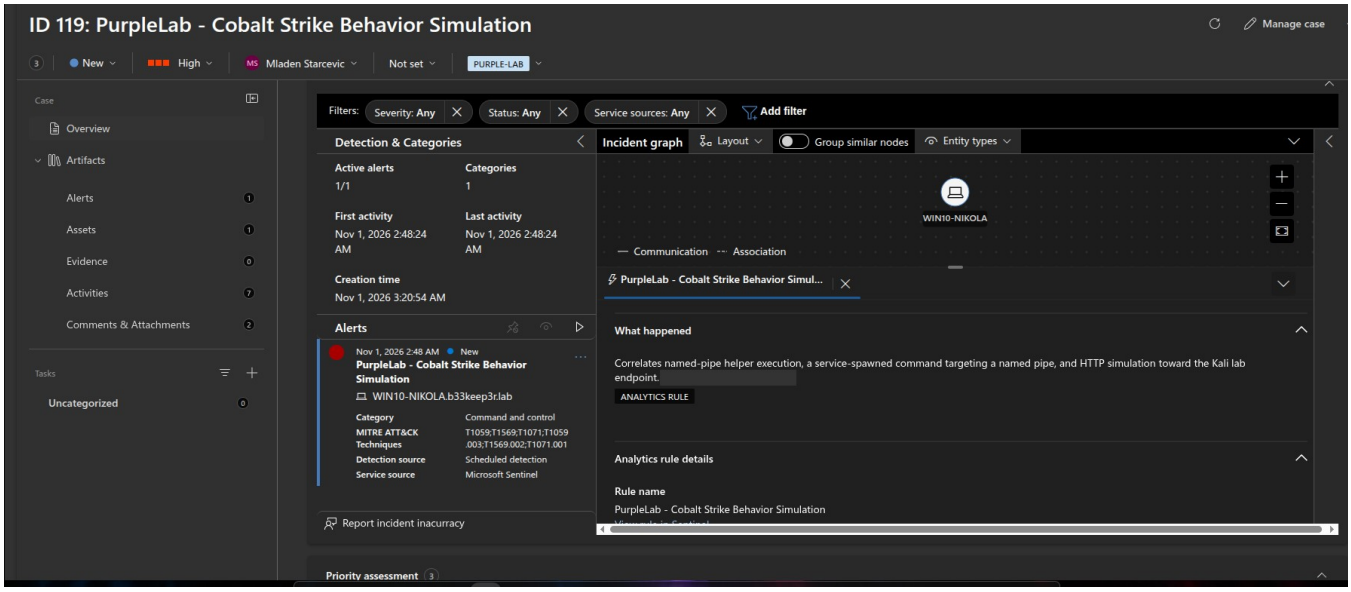
E06 The console shows service creation and removal, Access denied and a Defender notification.
Source: image(20261101-014857).png | cropped from original



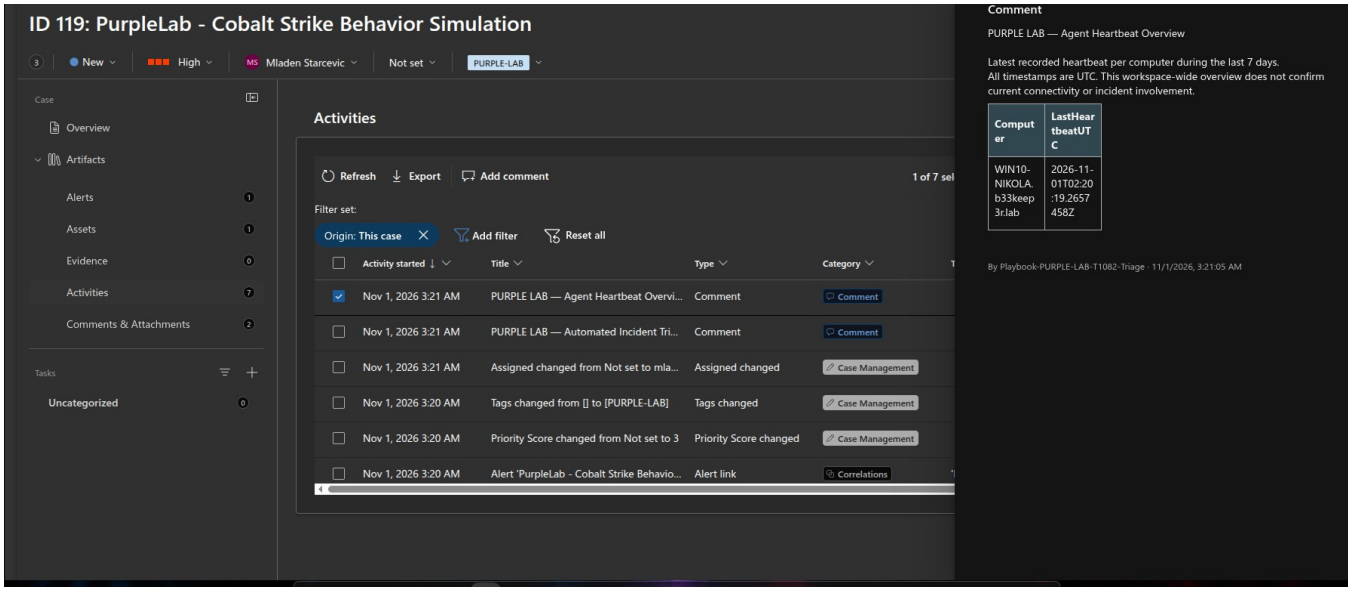
E07 Kali records HEAD requests from WIN10-NIKOLA for pixel.gif and a jquery resource; the server returns 404.
Source: image(20261101-014924).png

Result: Sentinel received 25 process events. Local event 7045 confirms the tbbd05 service, while Defender events 1116 and 1117 confirm the HackTool:Win32/NamedPipeImpers.A detection and the reported Remove action. The services.exe → cmd.exe process chain was recorded despite the console error.

Test 03 Incident and enrichment



E08 High incident 119 links the Cobalt LAB rule to WIN10-NIKOLA. The test tool name is omitted in the marked fields.
Source: image(20261101-022130).png | redacted



E09 Incident 119 has an owner, the PURPLE-LAB tag and two playbook comments: Heartbeat Overview and Automated Incident Triage.
Source: image(20261101-022139).png

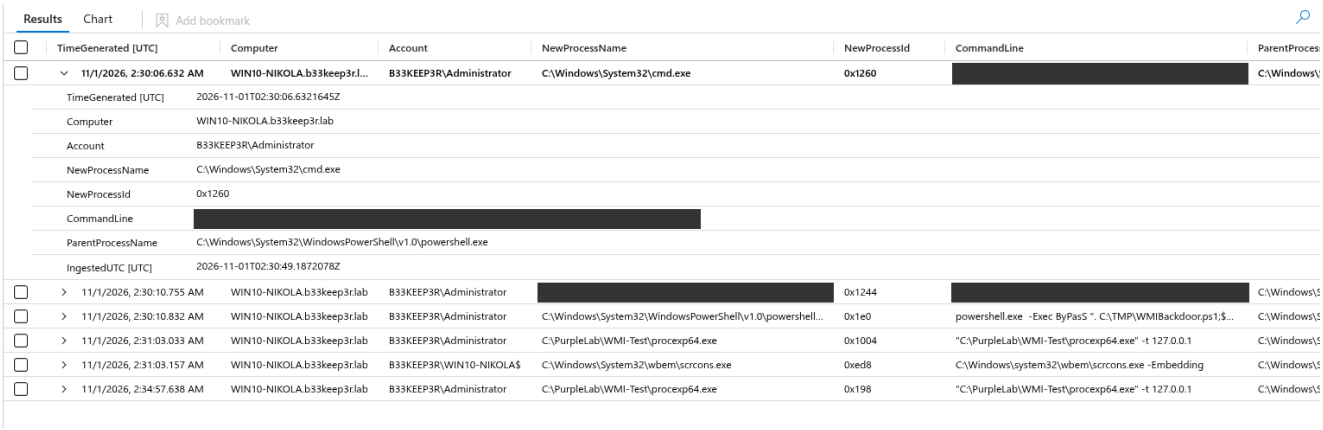
Result: incidents 117 and 119 came from evaluation of the same earlier test. Playbook execution in incident 119 is confirmed; its trigger and the missing Security Event Summary remain to be checked. Incident creation time is not a measure of real-time detection latency.

Limitation: successful impersonation or privilege escalation was not established. Curl command lines combining --data-binary with -I do not confirm content transfer. The service no longer existed after the test, but C:\TMP\CreateNamedPipe.exe and C:\Windows\b6a1458f396.exe remained at the last check; their subsequent removal was not documented.

Test 04 WMI registration and control

MITRE ATT&CK T1546.003 | Sentinel detection pending validation

The wmi-backdoor.bat module registered ProcessStartTrigger, the KillProcess ActiveScriptEventConsumer and their binding in root\subscription. A copy of Windows ping.exe named procexp64.exe was used as the trigger; the actual Process Explorer was not run.



E10 Sentinel event 4688 shows the registration PowerShell, the test process, scrcons.exe and the control run after cleanup. The test tool name is omitted in the marked fields.
Source: image(20261101-023552).png | redacted

Check	Confirmed result
Registration 02:30:11 UTC	Local Sysmon 19/20/21 and WMI 5861/5859.
Trigger PID 4100	Process exited within 10 s; scrcons.exe was recorded immediately after launch.
Cleanup 02:34:37 UTC	Filter and consumer removed; the remaining binding was removed through WMI commands.
Control PID 408	After cleanup, the process remained running for 10 s and was then stopped manually.
Ingestion and detection	Event 4688 is ingested; Sysmon 19/20/21 was not confirmed by Sentinel queries.

Conclusion: local registration and the control test strongly support operation of the WMI consumer. No Defender events 1116–1119 were found in the checked interval. The custom WMI rule and incident have not yet been validated.

Remaining validation

Confirm Sysmon 19/20/21 ingestion through the DCR, then validate the WMI rule and incident. The presence of local events does not confirm their ingestion into Sentinel.

Evidence E01–E10 covers the earlier tests. Times, PIDs and local control results are retained to correlate them with Sentinel records.

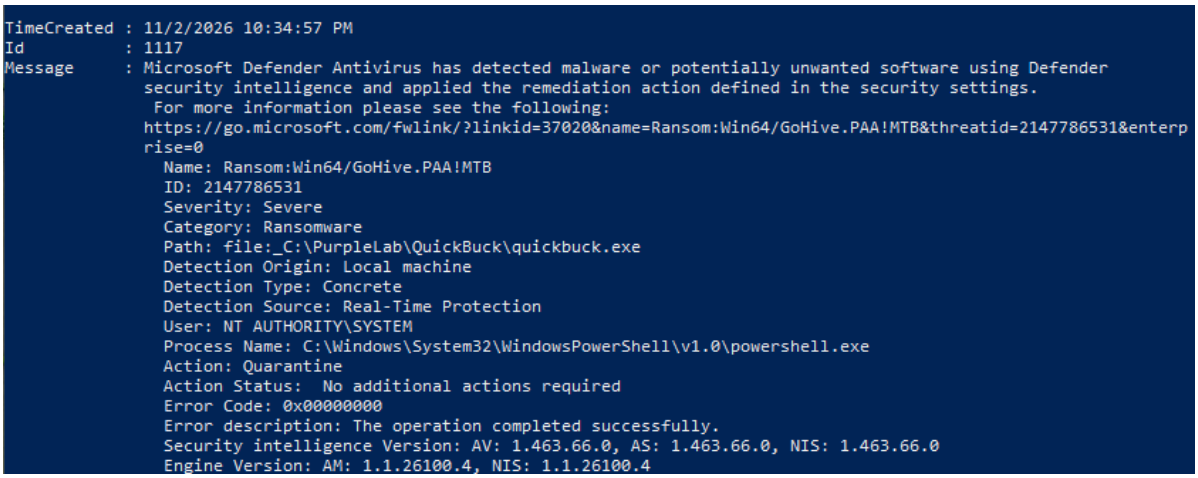
Test 05 QuickBuck protection validation

We tested AV prevention, execution against newly created test documents and triage capability. The Windows build was prepared on Kali and downloaded through the HTTP service at 10.0.2.8:8030. The verified SHA256 below links the build, downloaded EXE and local process record.

Executable SHA256

c202ed164ee6d984682d42c6a3b7e9f793a919151ce1e7ac04d390820f196501

During the initial downloads, Defender detected Ransom:Win64/GoHive.PAA!MTB and quarantined quickbuck.exe. Events 1116 and 1117 were also confirmed in the Log Analytics Event table. HTTP 200 on Kali confirms file delivery; successful execution requires separate process evidence.



E11 Defender event 1117 confirms successful quarantine of the original EXE and code 0x00000000. The console shows local time.
Source: image(20261102-213601).png | cropped from original

Phase on 2 November UTC	Confirmed outcome
21:58:39 → 21:58:55 22:21:19 → 22:21:34	Repeated detections and quarantine of the original EXE.
23:06:37 → 23:06:58	The EXE starts, but Defender blocks its WINWORD.EXE copy in System32. ExitCode 1; the documents and note do not exist.

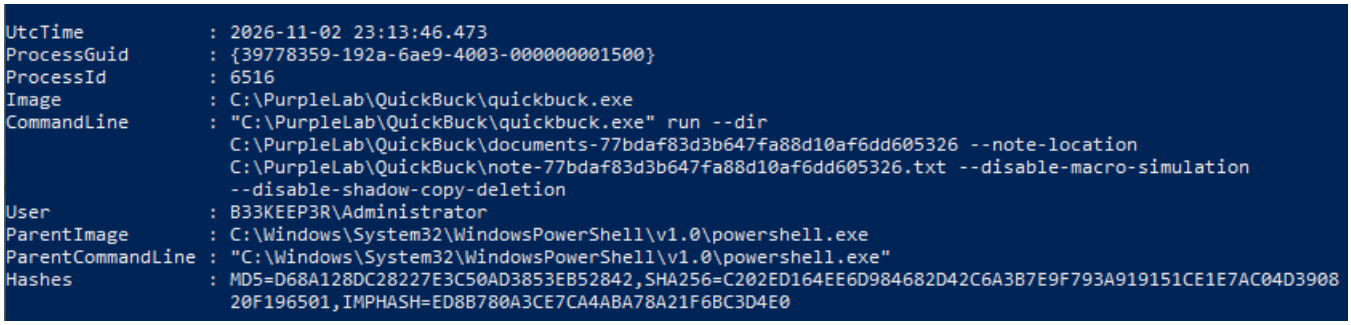
The first attempt used a temporary exclusion for quickbuck.exe only. The exclusion did not cover the WINWORD.EXE copy. This was a copy of the test EXE rather than Microsoft Word; its successful execution was not confirmed.

Before the final test, the Sysmon FileCreate include filter was extended to C:\PurpleLab\QuickBuck\. The configuration was validated, and a probe TXT file generated local event 11 under the PurpleLab-QuickBuck rule at 23:03:28.989 UTC. This resolved local filtering of the evidence files.

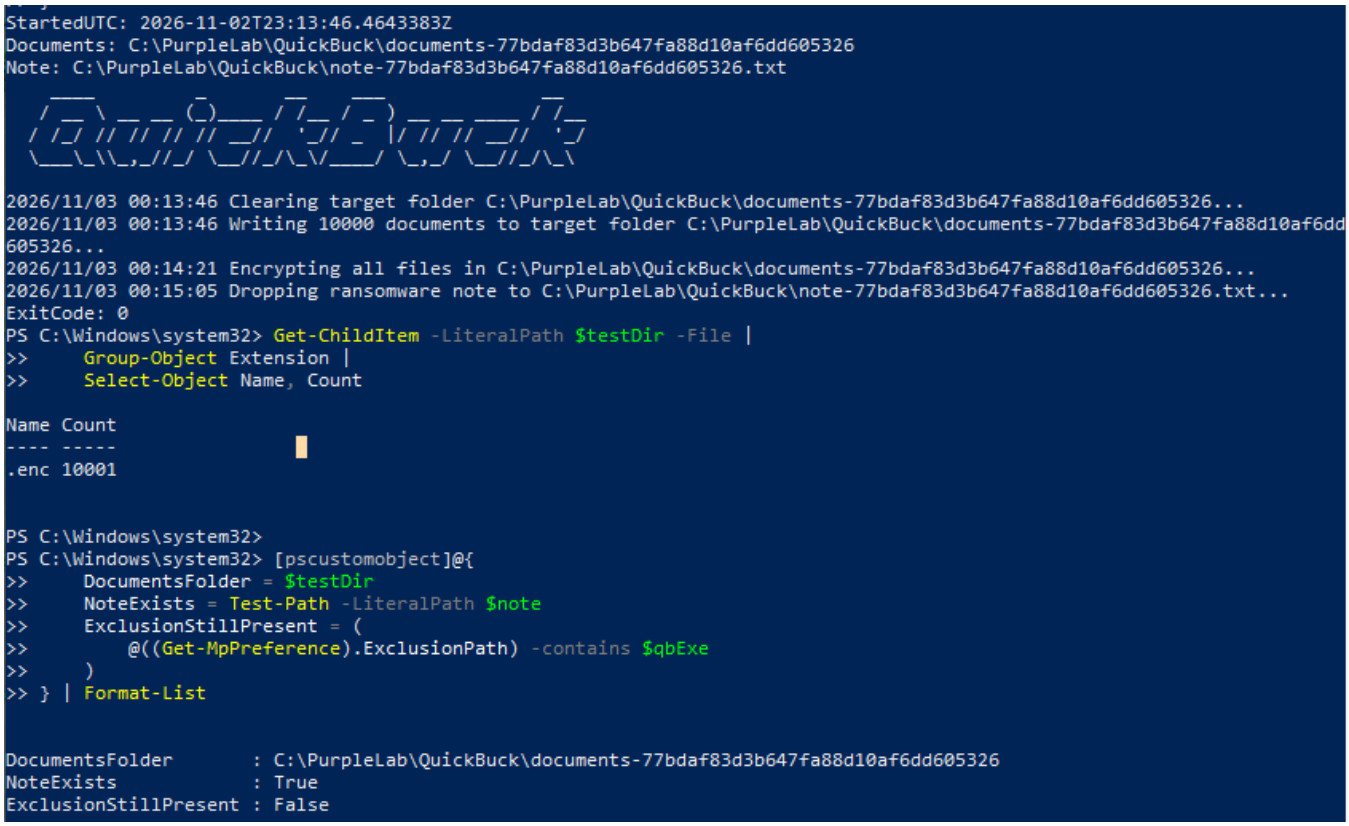
Test 05 Completed execution

The completed test started on 2 November 2026 at 23:13:46 UTC, or 3 November at 00:13:46 local time. A new documents-77bdaf83d3b647fa88d10af6dd605326 folder under C:\PurpleLab\QuickBuck\ was used. The folder did not exist before the test; the tool created the test documents inside it.

The run used --disable-macro-simulation and --disable-shadow-copy-deletion, with a temporary exclusion for the original EXE only. The Word/macro chain and shadow copy deletion were not tested in this run.



E12 Sysmon process record for the completed test: PID 6516, PowerShell parent, Administrator account, command line and hash.
Source: image(20261102-231718).png | cropped from original

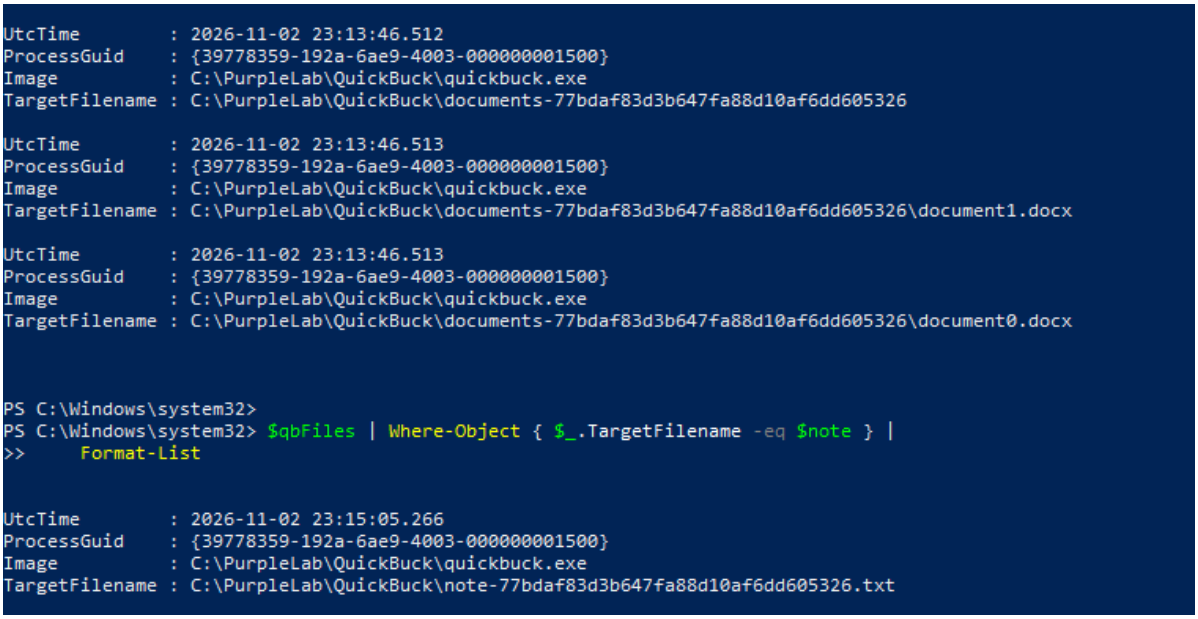


E13 The console shows completion with ExitCode 0, 10,001 .enc files, an existing note and the removed exclusion.
Source: image(20261102-231536).png | cropped from original

Result: the tool output, disk inventory and local Sysmon evidence support a successfully completed simulation of test document encryption. Cryptographic correctness of the content and recovery through decryption were not independently verified.

Test 05 Local triage and evidence

Process and file events were correlated using ProcessGuid {39778359-192a-6ae9-4003-000000001500}. The same process created the test documents, .enc files and note-77bdaf83d3b647fa88d10af6dd605326.txt.



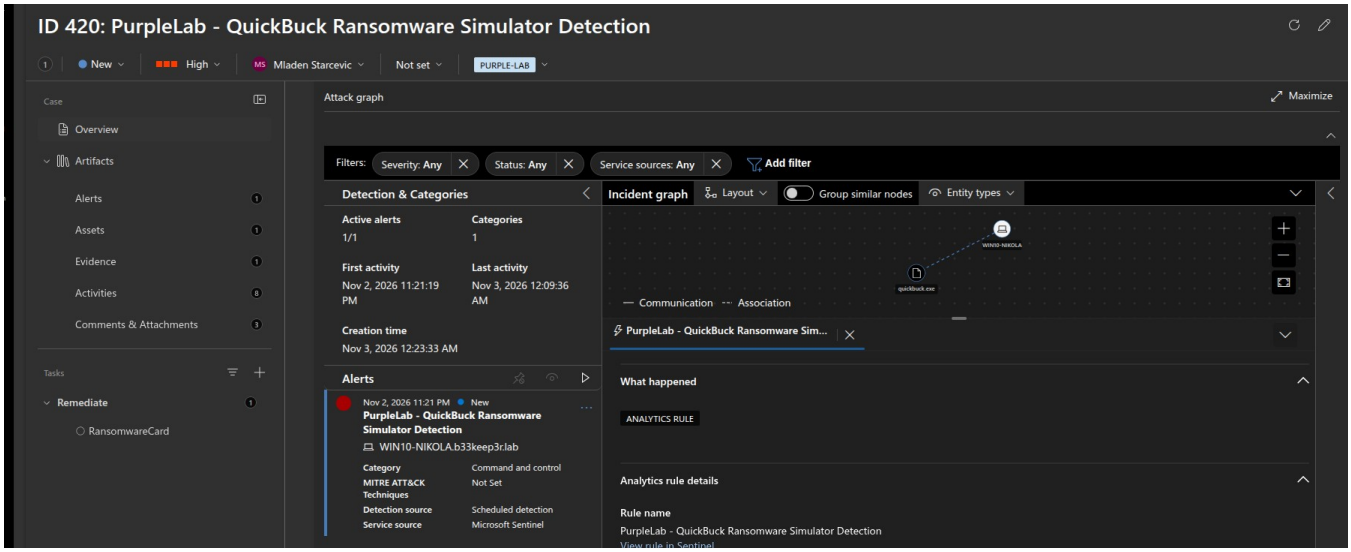
E14 Local Sysmon events link the initial documents and ransom note to the same process. The note was created at 23:15:05.266 UTC.
Source: image(20261102-232013).png | cropped from original

Metric	Value on 2 November UTC
Process start	23:13:46.473 | PID 6516
Total FileCreate events	20,004 for the correlated process
FileCreate events for .enc files	10,001 events; 10,001 .enc files in the inventory
First and last .enc	23:14:21.513 → 23:15:05.265
Note creation	23:15:05.266

The note claims data theft and disclosure and contains a fake onion address. This is test message content; the evidence does not confirm exfiltration.

SIEM coverage: the Event query for Sysmon during the completed test returned no data. Local events 1 and 11 are confirmed, while their ingestion into Sentinel and alerting on bulk .enc file creation remain to be validated.

Test 05 Sentinel and automation



E15 High incident 420 with one host and the quickbuck.exe file, an owner and the PURPLE-LAB tag. The portal shows local time.
Source: image(20261102-232501).png

The PurpleLab - QuickBuck Ransomware Simulator Detection rule processes Defender detections. In incident 410, the domain was mapped separately as a host. The correction combined HostName and DnsDomain within one Host entity, and Name and Directory within one File entity; incidents 413 and 420 show one host and one file.

Automation assigned the owner Mladen Starcevic and the PURPLE-LAB tag, and added triage, Heartbeat and Security Event Summary comments. For incident 420, the summary shows 250 events with ID 4688, 34 with ID 4624, and two each with IDs 4625 and 4648. These are host events within the enrichment window, not a count of QuickBuck actions. The last displayed heartbeat is 23:22:07.665 UTC.